



Malaria

Jeanne Rini Poespoprodjo, Nicholas M Douglas, Daniel Ansong, Steven Kho, Nicholas M Anstey

Lancet 2023; 402: 2328–45

Published Online

November 1, 2023

[https://doi.org/10.1016/S0140-6736\(23\)01249-7](https://doi.org/10.1016/S0140-6736(23)01249-7)

Centre for Child Health and
Department of Child Health,
Faculty of Medicine, Public
Health and Nursing,
Universitas Gadjah Mada,
Yogyakarta, Indonesia
(J R Poespoprodjo MD); Timika
Malaria Research Facility,
Papuan Health and Community
Development Foundation,
Timika, Indonesia
(J R Poespoprodjo, S Kho PhD);
Mimika District Hospital and
District Health Authority,
Timika, Indonesia
(J R Poespoprodjo); Global and
Tropical Health Division,
Menzies School of Health
Research, Charles Darwin
University, Darwin, NT,
Australia (J R Poespoprodjo,
N M Douglas FRACP, S Kho,
Prof N M Anstey FRACP);
Department of Infectious
Diseases, Christchurch
Hospital, Te Whatu Ora
Waitaha, Christchurch,
New Zealand (N M Douglas);
Department of Medicine,
University of Otago,
Christchurch, New Zealand
(N M Douglas); School of
Medicine and Dentistry,
Kwame Nkrumah University of
Science and Technology,
Kumasi, Ghana
(Prof D Ansong FWACP);
Department of Infectious
Diseases, Royal Darwin
Hospital, Darwin, NT, Australia
(Prof N M Anstey)

Correspondence to:
Dr Jeanne Rini Poespoprodjo,
Timika Malaria Research Facility,
Papuan Health and Community
Development Foundation,
Timika 99910, Indonesia
didot2266@yahoo.com

Malaria is resurging in many African and South American countries, exacerbated by COVID-19-related health service disruption. In 2021, there were an estimated 247 million malaria cases and 619 000 deaths in 84 endemic countries. *Plasmodium falciparum* strains partly resistant to artemisinins are entrenched in the Greater Mekong region and have emerged in Africa, while *Anopheles* mosquito vectors continue to evolve physiological and behavioural resistance to insecticides. Elimination of *Plasmodium vivax* malaria is hindered by impractical and potentially toxic antirelapse regimens. Parasitological diagnosis and treatment with oral or parenteral artemisinin-based therapy is the mainstay of patient management. Timely blood transfusion, renal replacement therapy, and restrictive fluid therapy can improve survival in severe malaria. Rigorous use of intermittent preventive treatment in pregnancy and infancy and seasonal chemoprevention, potentially combined with pre-erythrocytic vaccines endorsed by WHO in 2021 and 2023, can substantially reduce malaria morbidity. Improved surveillance, better access to effective treatment, more labour-efficient vector control, continued drug development, targeted mass drug administration, and sustained political commitment are required to achieve targets for malaria reduction by the end of this decade.

Introduction

Malaria is a tropical and subtropical, mosquito-borne parasitic disease that is endemic in 84 countries and causes an estimated 247 million clinical infections and 619 000 deaths annually.¹ The infectious agent is a single-celled intracellular protozoan from the genus *Plasmodium*. Six species commonly infect humans (figure 1), of which *Plasmodium falciparum* and *Plasmodium vivax* are the most important. Morbidity and mortality from malaria stem from parasitic invasion of erythrocytes leading to haemolysis and, in *P falciparum* infections, sequestration of infected red cells in the microvasculature of vital organs.⁵ Most deaths from malaria are caused by *P falciparum* in sub-Saharan Africa, whereas *P vivax* is more prevalent in Asia and South America and predominantly causes anaemia from relapsing infections.⁶

Early diagnosis and prompt effective treatment of malaria prevents severe sequelae and death and reduces the risk of onward transmission of the parasite. Prevention strategies for malaria include various forms of chemoprophylaxis, mosquito vector control, and, in some regions of sub-Saharan Africa, vaccination. These strategies are continually threatened by evolutionary adaptation of the parasite and mosquito vector.

In the early 2000s, the large-scale introduction of artemisinin-based combination therapies (ACTs) for

uncomplicated malaria and intravenous artesunate for severe malaria and improved parasitological diagnosis contributed to a substantial decline in global malaria mortality.^{1,7,8} Unfortunately, global malaria incidence has increased in the past 5 years and new challenges, such as spreading artemisinin resistance and COVID-19-related disruption to health systems threaten malaria control.^{1,9,10}

Epidemiology

Malaria is transmitted by female *Anopheles* mosquitoes that breed in freshwater bodies in warm and humid environments.¹¹ Endemicity is determined by the geographical distribution of vector species, their proximity to humans, and the capacity of human populations to control vectors and effectively manage infection.¹² Therefore, malaria flourishes in low-income tropical regions and is exacerbated by social disruption, such as war or natural disaster.

Of an estimated 247 million malaria cases annually, 234 million (95%) occur in malaria-endemic countries in Africa where 99% of infections are caused by *P falciparum* (figure 2).¹ Four African countries were estimated to account for over half of all malaria deaths in 2021: Nigeria (31%), Democratic Republic of the Congo (13%), Niger (4%), and Tanzania (4%).¹ 7·4 million malaria cases occur in South America and the WHO South-East Asian and Western Pacific regions, where *P falciparum* and *P vivax* are often coendemic.¹ In 2021, WHO estimated there were 4·9 million *P vivax* cases.¹ The other three human-only *Plasmodium* species account for fewer than 0·5 million cases. *Plasmodium malariae* and the two non-recombining sympatric species *Plasmodium ovale curtisi* and *Plasmodium ovale wallikeri* have broad geographical distributions but due to difficulties in diagnosis, low parasitaemia, and relatively lower virulence, are commonly diagnosed incidentally.¹³ *Plasmodium knowlesi* infects macaque monkeys in southeast Asia¹⁴ and causes zoonotic, sometimes fatal, malaria in humans.¹⁵ Seven other zoonotic species reported

Search strategy and selection criteria

We searched PubMed, MEDLINE, Web of Science, Embase, and the Cochrane Database of Systematic Reviews using the search term “malaria”. We were unable to include all references on this topic, and so we prioritised clinical trials, systematic reviews, and studies of malaria pathogenesis or epidemiology that were published in English between Jan 1, 2018, and Jan 31, 2023. We also screened the reference lists of recently published articles for older seminal studies and reviewed guidance documents in WHO’s online repository.

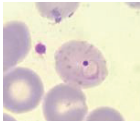
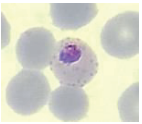
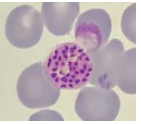
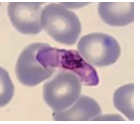
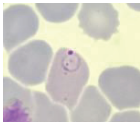
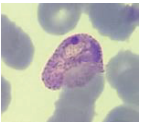
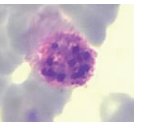
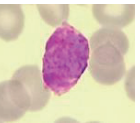
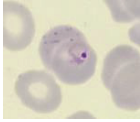
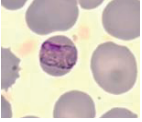
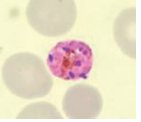
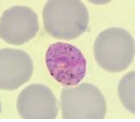
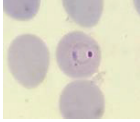
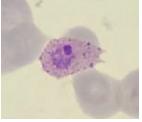
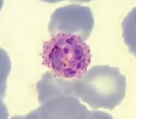
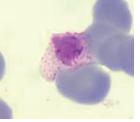
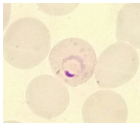
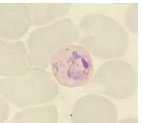
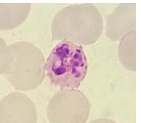
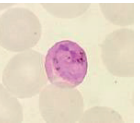
Human-only species					Characteristics
	Rings	Trophozoites	Schizonts	Gametocytes	
<i>Plasmodium falciparum</i>					• Usually higher parasitaemia compared with other species • Maurer's clefts can be visible in rings and trophozoites • Trophozoites, schizonts, and immature gametocytes not normally seen in blood smears (sequester and accumulate in microvasculature and tissues) • Schizonts comprise 8–24 merozoites • Crescent-shape characteristic of mature gametocytes
<i>Plasmodium vivax</i>					• Larger cytoplasm in rings; Schüffner's dots can be visible • Trophozoites amoeboid-shaped; irregular cytoplasm • Schizonts enlarged and comprise 12–24 merozoites • Round-to-oval-shaped gametocytes; scattered pigment • All stages present in blood smears (most biomass present in several organs, predominantly the spleen) • Strict tropism for infecting immature (CD71+) reticulocytes
<i>Plasmodium malariae</i>					• Parasitised red cells not enlarged • Rings sometimes small with large chromatin dot • Trophozoites have compact cytoplasm; occasionally present as band forms; coarse, dark pigment • Schizonts comprise 6–12 merozoites around dark pigment mass; merozoites occasionally form rosettes • All stages present in blood smears
<i>Plasmodium ovale wallikeri</i> and <i>Plasmodium ovale curtisi</i>					• Robust cytoplasm in rings; Schüffner's dots can be visible • Trophozoites round to oval, often tear-drop shaped; compact cytoplasm with large chromatin dot • Schizonts comprise 6–14 merozoites around pigment mass • All stages present in blood smears • Strict tropism for infecting immature (CD71+) reticulocytes • Occasionally misidentified as <i>P. vivax</i>
Zoonotic species					
<i>Plasmodium knowlesi</i>					• Rings resemble <i>P. falciparum</i> • Trophozoites have compact cytoplasm; occasionally present as band form; resemble <i>P. malariae</i> • Schizonts comprise up to 16 merozoites around dark pigment mass; merozoites occasionally form rosettes • All stages present in blood smears • Geographical distribution: southeast Asia
	Characteristics				Geographical distribution in humans
<i>Plasmodium cynomolgi</i>	Morphologically similar to <i>P. vivax</i>				Southeast Asia
<i>Plasmodium inui</i>	Morphologically similar to <i>P. malariae</i>				Malaysia and Thailand
<i>Plasmodium coatneyi</i>	Morphologically similar to <i>P. falciparum</i>				Malaysia
<i>Plasmodium semiovale</i>	Morphologically similar to <i>P. ovale</i>				Malaysia
<i>Plasmodium simium</i>	Morphologically and molecularly similar to <i>P. vivax</i>				Brazil
<i>Plasmodium brasilianum</i>	Morphologically and molecularly similar to <i>P. malariae</i>				Central and South America
<i>Plasmodium fieldi</i>	Morphologically similar to <i>P. vivax</i> and <i>P. ovale</i>				Thailand (co-infection)

Figure 1: Microscopy of *Plasmodium* species infecting humans

Geographical distribution of *P. cynomolgi*, *P. inui*, *P. semiovale*, *P. coatneyi*, *P. simium*, and *P. brasilianum* described in Antinori et al.² and Yap et al.³ Cryptic *P. fieldi* infections recently discovered in Thailand.⁴

to cause geographically restricted human infections are rare (figure 1).

The intensity of malaria transmission determines the age distribution of clinical malaria and malaria-associated mortality.¹⁶ In many parts of sub-Saharan Africa, individuals receive more than ten infectious mosquito bites per year.¹⁷ In these environments, intense parasite exposure leads to rapid development of premunition (protection against high parasitaemia and illness without

eliminating the infection), skewing the burden of severe malaria to children younger than 5 years.⁵ In such highly endemic settings, severe malaria in adults is unusual, but a substantial proportion of the population will have asymptomatic, submicroscopic parasitaemia.⁵ Asymptomatic parasitaemia is a major contributor to malaria transmission.¹⁸

Outside of Africa, transmission of *P. falciparum* and *P. vivax* is generally lower with many areas having

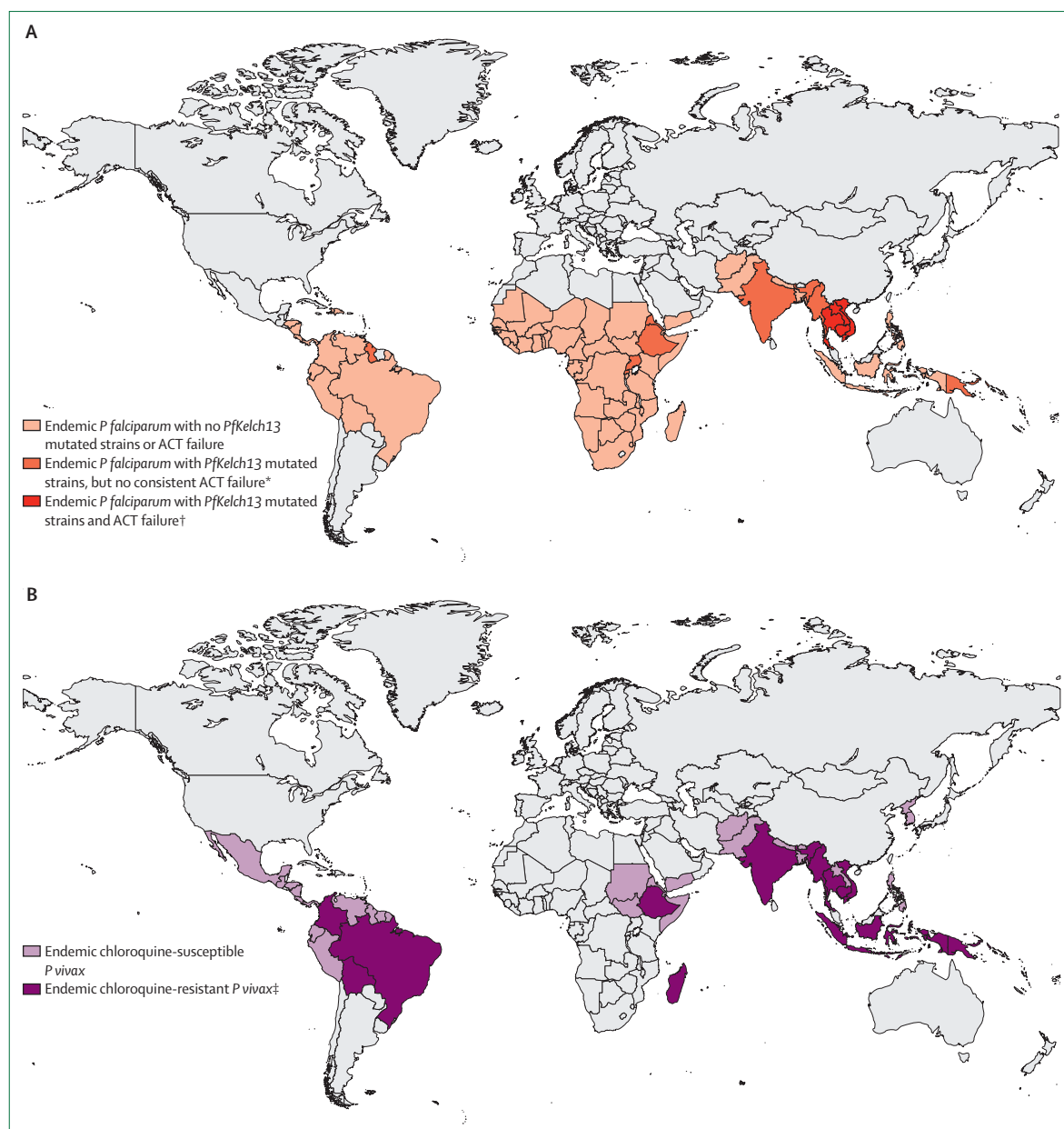


Figure 2: Map of *Plasmodium falciparum* (A) and *Plasmodium vivax* (B) endemicity and antimalarial drug resistance as at Jan 31, 2023

ACT=artemisinin-based combination therapies. *≥5% of isolates in one or more provinces positive for one or more WHO-validated or WHO-associated *PfKelch13* mutations associated with partial artemisinin resistance. †≥5% of isolates in one or more provinces positive for one or more WHO-validated or WHO-associated *PfKelch13* mutations associated with partial artemisinin resistance and consistent day-28 failure rate of ≥10% after artemether-lumefantrine, dihydroartemisinin-piperaquine, or artesunate-mefloquine. ‡At least one study showing day-28 failure ≥10% after chloroquine treatment. Figure created with mapchart.net.

seasonal or sporadic incidence.¹⁹ In these lower-endemicity environments, all age groups are at risk of severe *falciparum* malaria.¹⁹ Submicroscopic, asymptomatic parasitaemia is also common. Premunition against *P. vivax* malaria tends to develop earlier than for *P. falciparum* in coendemic locations.^{19,20} Non-immune travellers and people displaced into endemic regions are at high risk of severe malaria at any age.⁵

Biology

Sporozoites from female *Anopheles* mosquitoes enter the human circulation following a blood meal and are transported to the liver to invade hepatocytes (figure 3). After 5–8 days of pre-erythrocytic incubation (approximately 15 days for *P. malariae*²¹), merozoites are released from rupturing hepatic schizonts and enter circulation to invade red blood cells, initiating the intraerythrocytic lifecycle responsible for malaria

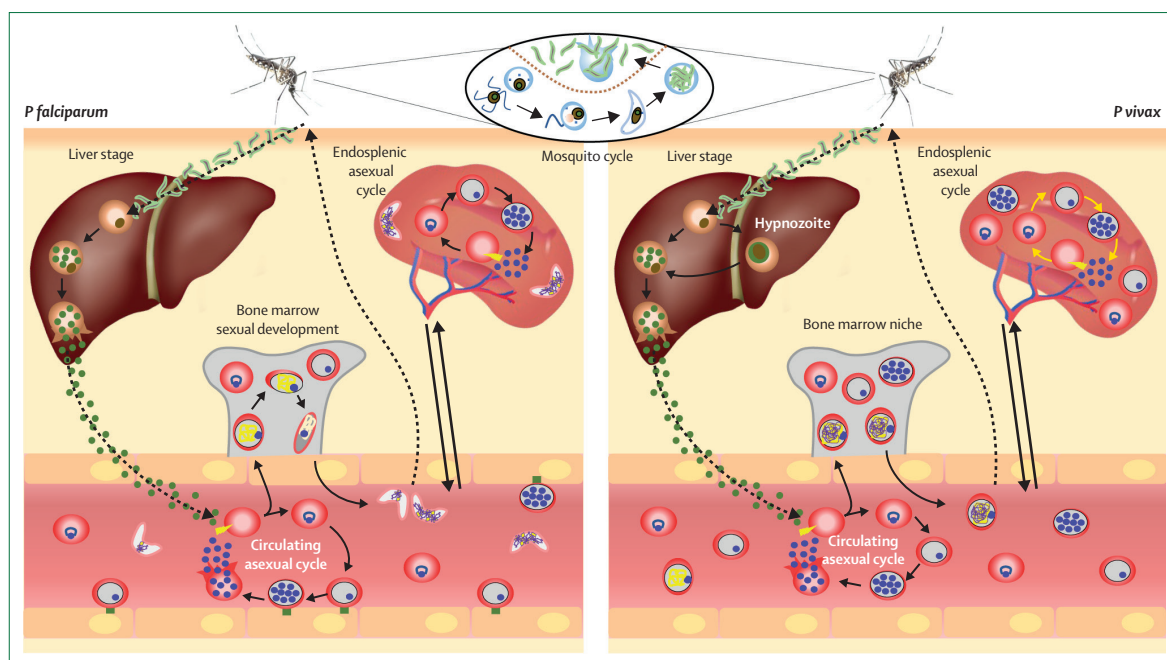


Figure 3: *Plasmodium falciparum* and *Plasmodium vivax* lifecycle

Human infection begins when a mosquito inoculates sporozoites into the skin during a blood meal. Sporozoites subsequently infect hepatocytes. Pre-erythrocytic incubation ends when hepatic schizonts release merozoites (green dots) from the liver into the bloodstream to commence the intraerythrocytic asexual lifecycle. Asexual stages (rings, trophozoites, and schizonts) undergo maturation within erythrocytes followed by erythrocyte rupture, with subsequent reinvasion of merozoites (purple dots) into uninfected erythrocytes in the spleen (greater for *P. vivax* than *P. falciparum*) and peripheral blood. Sexual-stage development occurs in the bone marrow. Mature gametocytes are released into the bloodstream and are subsequently ingested by a mosquito during a blood meal, facilitating onward parasite transmission.

symptoms. Asexual parasite development in erythrocytes occurs over 24–72 h depending on species, progressing from rings, to trophozoites, and finally to multinucleated erythrocytic schizonts (figure 1). At schizogony, merozoites released into the bloodstream reinvade surrounding erythrocytes, thus commencing a new asexual cycle. Although *P. falciparum* invades red blood cells of all ages, *P. vivax* and *P. ovale* only invade young reticulocytes expressing CD71.²² During physiological splenic passage, asexual stages of *P. falciparum* and *P. vivax* are retained in the spleen initiating an endosplenic lifecycle in both species.²³ The reticulocyte-rich spleen sustains the majority of asexual-stage biomass in *P. vivax* infections.²⁴ Asexual parasites are also found in the bone marrow in falciparum and vivax infections.^{25,26} Multiple cycles of intravascular asexual replication allow the parasite to proliferate rapidly and drive pathogenesis. Individuals infected with *P. vivax* or *P. ovale* develop latent hypnozoites within hepatocytes, which can activate weeks or months after the initial infection, causing relapsing blood-stage infection.

The bone marrow is a major site for the development of sexual stages, whereby sexually committed rings differentiate into gametocytes following environmental and host cues favouring transmission. After 9–12 days, mature *P. falciparum* gametocytes enter circulation and become infectious after a further 2–3 days.²⁷ *P. vivax*

gametocytes mature faster (2–3 days) and infectious gametocytes are typically found in circulation before the onset of symptoms. After gametocyte ingestion by a competent mosquito, sexual reproduction occurs in the mid-gut to form a zygote. Gut ookinets transform into an oocyst containing thousands of sporozoites, which migrate to salivary glands for onward transmission.

Pathogenesis

P. falciparum

Severe falciparum malaria is caused by microvascular sequestration and obstruction from parasitised red cells cytoadhering to activated and dysfunctional endothelial cells.^{5,28} Sequestered parasite biomass and endothelial activation are each independently associated with impaired perfusion, severity of organ dysfunction, and mortality.^{29,30} Cytoadherence is mediated by binding of the PfEMP1 family of proteins expressed on parasitised erythrocytes to receptors on endothelial cells and other red cells. Different PfEMP1 variants encoded by var genes have different binding affinities to receptors differentially expressed in different organs, thereby contributing to risk of specific severe malaria manifestations.^{31,32} Endothelial receptors include EPCR, ICAM-1, and CD36. PfEMP1 variants with increased binding to EPCR, especially dual binding to EPCR and ICAM-1, are associated with cerebral malaria. These variants mediate not only sequestration of

parasitised red cells,^{31,33} but also loss of EPCR-dependent cytoprotection,^{28,34} including localised fibrin deposition, platelet activation, loss of endothelial barrier-function, and microhaemorrhages. Endothelial dysfunction in malaria is exacerbated by degradation of cytoprotective glycocalyx, reduced bioavailability of nitric oxide, Weibel-Palade body exocytosis, release of free haem and parasite histones from red cells, and release of neutrophil extracellular traps, with each of these processes associated with disease severity.^{28,29,35–37}

Autopsy studies in cerebral malaria show strong associations between depth of coma, the magnitude of sequestration of parasitised red cells, and vascular congestion with both infected and uninfected red cells.³⁸ In children with cerebral malaria, brain swelling is the central radiological finding^{39,40} and is strongly associated with mortality, suggesting raised intracranial pressure and brainstem herniation are on the causal pathway to death in this age group.³⁹ In addition to vascular congestion, other proposed causes of brain swelling include vasogenic oedema and haemorrhagic breaches in the blood–brain barrier, autoregulatory dysfunction, and, to a lesser degree, cytotoxicity.^{38,39,41,42} In a subset with appearances similar to posterior reversible encephalopathy syndrome, blood–brain barrier dysfunction, venous congestion, and autoregulatory dysfunction might be causative.⁴³ Retinal studies suggest that hypoxic injury underlies neurological sequelae in survivors.^{42–44} In adult cerebral malaria, brain swelling is less severe and is not associated with mortality.^{40,45}

Major contributors to metabolic acidosis are lactate from inadequate tissue oxygenation and impaired clearance,⁵ other host organic acids,⁴⁶ and gut microbial acids that translocate into the bloodstream.⁴⁷ This increased gut permeability arises from parasite sequestration⁴⁸ and impaired production of gut-protective citrulline and glutamine,⁴⁹ and contributes to the increased risk of concomitant bacteraemia in severe malaria with *Salmonella* and other Gram-negative bacteria.⁵⁰ Shock is less common in severe malaria than in severe sepsis, with increased systemic vascular resistance in malaria arising from nitric oxide scavenging by haemolysis-related cell-free haemoglobin.⁵¹ Relative hypovolaemia and reduced cardiac index reserve are found in patients before death.⁵² Acute kidney injury (AKI) is associated with parasite sequestration, free haem-mediated oxidative damage to renal tubular epithelium, glycocalyx degradation, and endothelial activation.^{53–55}

Anaemia in acute malaria is multifactorial. Loss of uninfected red cells exceeds the destruction of parasitised red cells in all species.⁵⁶ Oxidative damage, G6PD deficiency, complement-mediated destruction, reduced red cell deformability, and increased splenic retention and clearance all contribute to loss of uninfected red cells.^{23,56,57} Dyserythropoiesis impairs erythrocyte production.⁵⁶ Anaemia is exacerbated by prolonged parasitaemia and delayed presentation^{58,59} and parasite clearance.⁵⁹ Thrombocytopenia is also

multifactorial, but a major cause is platelet-binding to infected red cells.⁶⁰

Placental malaria is caused by sequestration of parasitised red cells in the intervillous spaces, mediated by *P. falciparum* VAR2CSA binding to placental chondroitin sulphate A⁶¹ with associated monocyte infiltration,⁶² angiopoietin 1 and 2 dysregulation, and vascular disruption.^{62,63} Malaria in early pregnancy (ie, <15 weeks gestational age) alters placental angiogenesis, reduces uteroplacental perfusion, and impairs fetal growth.⁶⁴

P vivax

There is minimal microvascular sequestration in vivax malaria, but greater extravascular parasite accumulation in spleen and bone marrow.^{23–25} Circulating asexual-stage infected red cells are substantially more deformable than *P. falciparum*.^{65,66} Although less common than in falciparum malaria, organ dysfunction is associated with parasite biomass-related endothelial activation and systemic inflammation.^{67,68} Severe anaemia is associated with repeated relapses from liver hypnozoites before haematological recovery, causing progressive loss of both infected and uninfected red cells, with associated mortality.^{69–71} Chronic submicroscopic infections are sustained by both hypnozoite relapses and release from the extravascular splenic reservoir.²³ In the placenta, there is no parasite sequestration, but placental insufficiency is associated with inflammatory responses, dysregulation of angiopoietin 1 and 2, vascular disruption, and maternal anaemia.^{72,73}

P knowlesi

Fatal knowlesi malaria is associated with microvascular accumulation of parasitised red cells through mechanisms different to those in *P. falciparum*.⁷⁴ Impaired perfusion, organ injury, and severe disease are also associated with reduced deformability of uninfected and infected erythrocytes, endothelial activation, glycocalyx degradation, and haemolysis.^{74–76}

Clinical presentation

Malaria is a non-specific febrile illness that cannot be differentiated from other infections on clinical grounds alone. Typical symptoms include fever, chills, headache, vomiting, diarrhoea, and cough. In highly endemic areas, parasitaemia is often coincidental to an alternative cause of illness.⁷⁷ Parasitaemia, platelet count, and plasma HRP2 thresholds for falciparum malaria have been evaluated in regions of different endemicity to distinguish the likelihood of malaria causing symptoms and to differentiate uncomplicated from severe malaria.^{77–79} In non-endemic regions, a travel history is essential to ensure malaria is considered in the differential diagnosis and excluded.

Clinical features of severe falciparum malaria vary with age and pregnancy status.^{5,80} Signs of severe malaria in children in endemic areas include prostration, reduced consciousness, respiratory distress, severe anaemia,

hypoglycaemia, shock, and jaundice (panel 1).^{77,82} Severe AKI, acute respiratory distress syndrome (often commencing after treatment), and jaundice are more common in adults than children, whereas severe anaemia and convulsions are generally less common.⁵ Pregnant women are particularly susceptible to hypoglycaemia, severe anaemia, and jaundice.⁸³ Respiratory distress in malaria is associated with severe anaemia and metabolic acidosis.^{77,82} Abnormal posturing and prolonged and repeated convulsions commonly accompany coma and can cause long-term neurological sequelae.^{5,77} Blackwater fever is commonly reported in African children.⁸² AKI is now recognised as an important complication in paediatric as well as adult severe malaria and is associated with risk of death.⁸⁴

WHO describes epidemiological and research criteria for severe malaria (panel 1).^{5,81} In routine clinical practice, clinicians should worry less about epidemiological and research definitions of severity: a patient who is unable to take oral medications or has organ dysfunction or high parasitaemia (>2%) is at increased risk of dying and should be treated for severe malaria with parenteral therapy and the best supportive care available.⁸¹

Within the broad definition of severe malaria, some syndromes are associated with lower in-hospital mortality rates (eg, severe anaemia) and others with higher mortality rates (eg, acidosis).⁸¹ In acute falciparum malaria, the risk of in-hospital death rises with a haemoglobin less than 5 g/dL, markedly so if less than 3 g/dL.⁸⁵ The risk for death increases in the presence of multiple complications. Impaired consciousness, base deficit (base excess less than -8 mmol/L), elevated blood urea (blood urea nitrogen ≥ 20 mg/dL), and clinical signs of underlying chronic illness (eg, lymphadenopathy) are independent predictors of death in children, with more than one complication increasing risk of mortality from 6% to 43%.^{8,86} A simple bedside score can predict risk of death in adults with severe malaria.⁸⁷

The clinical features of uncomplicated vivax malaria are similar to falciparum malaria. Severe manifestations of infection are less common than in falciparum malaria except for severe anaemia in infants and young children in high-transmission areas, such as the island of New Guinea.^{85,88,89} In these regions, relapses occur every 3–6 weeks, accounting for 60–80% of clinically apparent *P. vivax* infections.^{88–91} Failure to prevent relapses with hypnozoitocidal therapy can lead to progressive chronic, severe anaemia and delayed mortality.^{70,88,92} Organ dysfunction is less common in vivax malaria than falciparum malaria, but can include acute lung injury, hypotension, AKI,^{67,93} and, rarely, coma.⁹⁴ Severe *P. vivax* malaria in very low transmission settings is rare.

Uncomplicated and severe knowlesi malaria presents similarly to falciparum malaria, except that coma is notably absent and over 90% of patients are adults.^{15,74} Although patients can develop hyperparasitaemia, severe and fatal disease occurs at lower parasitaemia than in falciparum

Panel 1: WHO epidemiological and research criteria for severe malaria⁵

- Impaired consciousness: a Glasgow coma score <11 or a Blantyre coma score <3 in children younger than 5 years or who have not learned to speak
- Pulmonary oedema: radiologically confirmed or oxygen saturation <92% on room air with a respiratory rate >30 per min, often with chest indrawing and crepitations on auscultation
- Substantial bleeding: including recurrent or prolonged bleeding from the nose, gums, or venepuncture sites; haematemesis or melaena
- Shock: compensated shock is defined as capillary refill ≥ 3 s without hypotension. Decompensated shock is defined as systolic blood pressure <70 mm Hg in children or <80 mm Hg in adults, with evidence of impaired perfusion (ie, cool peripheries or prolonged capillary refill)
- Acidosis: a base deficit of >8 mEq/L, a plasma bicarbonate concentration of <15 mmol/L, or venous plasma lactate ≥ 5 mmol/L. Severe acidosis manifests clinically as respiratory distress (ie, rapid, deep, and laboured breathing)
- Hypoglycaemia: blood or plasma glucose <2.2 mmol/L (<40 mg/dL)
- Severe malarial anaemia: haemoglobin concentration ≤ 5 g/dL or a haematocrit of $\leq 15\%$ in children aged <12 years (haemoglobin concentration <7 g/dL and haematocrit <20% in individuals aged ≥ 12 years) with a parasite count >10 000 per μ L
- Severe acute kidney injury: plasma or serum creatinine >265 μ mol/L (3 mg/dL) or blood urea >20 mmol/L in individuals aged >12 years; not defined by WHO in children aged <12 years
- Jaundice: plasma or serum bilirubin >50 μ mol/L (3 mg/dL) with a parasite count >100 000 per μ L; *Plasmodium knowlesi*: jaundice and parasite density >20 000 per μ L
- Hyperparasitaemia: *Plasmodium falciparum* parasitaemia >10% of infected erythrocytes in stable high endemicity area; *P. knowlesi* parasite density >100 000 per μ L

Additional epidemiological criteria for severe malaria included in WHO malaria guidelines:⁸¹

- Prostration: the person is unable to sit, stand, or walk without assistance
- Multiple convulsions: more than two episodes within 24 h
- Severe vivax malaria: defined as for falciparum malaria but with no parasite density thresholds

Clinical criteria for use of intravenous artesunate

- Unable to take oral medications, or
- any research or epidemiological criteria for severe malaria, or
- any evidence of dysfunction of vital organs, or
- high parasitaemia (for *P. falciparum*: $\geq 2\%$ in low-transmission regions and in non-immune travellers and $\geq 4\%$ in high-transmission regions; for *P. knowlesi*: >15 000 parasites per μ L)

malaria.^{15,74,95} *P. malariae* causes chronic low-parasitaemia infections and anaemia.⁸⁵ Nephrotic syndrome can occur as a late sequela of *P. malariae* infection in approximately 1 in 500 microscopy-positive patients.⁹⁶

Laboratory diagnosis

Microscopy of thick and thin blood films remains the gold standard for diagnosing malaria. This method can differentiate *Plasmodium* species and quantify parasitaemia, both important for diagnosing and prognosticating severe malaria, targeting appropriate treatment, and monitoring therapeutic response.⁹⁷ Unless performed by experienced technicians, microscopy results might not be reliable.⁹⁷

Malaria rapid diagnostic tests (RDTs) are used as an alternative or adjunct to microscopy, particularly in settings with limited access to quality microscopy. RDTs that detect HRP2, an antigen exclusively produced by *P falciparum*, are usually sensitive for diagnosis of this species, but cannot be used to evaluate treatment response because HRP2 can persist in once-infected, parasite-free erythrocytes after splenic pitting.⁹⁸ False-negative results arise from *P falciparum* isolates without either or both of the PfHRP2 and PfHRP3 genes, increasingly reported across South America, Africa, and Asia.^{98–100} These gene deletions threaten the utility of HRP2-based diagnostics.⁹⁸

Non-falciparum malaria species can be detected with pLDH RDTs,⁹⁷ including *P knowlesi*,¹⁰¹ although cross-reactivity of species-specific pLDH remains a barrier for accurate RDT-based diagnosis in regions coendemic for

multiple non-falciparum infections, including zoonotic species.¹⁰²

Nucleic acid testing (NAT), including as part of multiplex fever panels, is highly sensitive¹⁰³ and useful in reference laboratories. Although used in some non-endemic settings for acute diagnosis in undifferentiated fever, NAT is unable to quantitate parasitaemia to guide use of parenteral therapy, and in most endemic settings is not available or ideal for immediate treatment decisions.

Case management

Comprehensive guidelines for management of severe and uncomplicated malaria are provided by WHO.^{5,81}

Severe malaria

All patients, regardless of infecting *Plasmodium* species (figure 1) and pregnancy status, who fulfil criteria for

Panel 2: Malaria treatment⁸¹

Severe malaria

First line

- Intravenous artesunate: 2.4 mg/kg per dose at 0 h, 12 h, and 24 h, then every 24 h if necessary; if bodyweight is <20.0 kg, give 3.0 mg/kg per dose to ensure equivalent exposure to the drug

Alternatives if artesunate not available

- Intramuscular artemether: initial dose 3.2 mg/kg, then 1.6 mg/kg every 24 h
- Intravenous quinine diluted in 5% dextrose: loading dose of 20 mg/kg infused over 4 h, then 10 mg/kg every 8 h infused no faster than 5 mg/kg per h
- Pre-referral rectal artesunate: recommended in primary health-care settings in which parenteral drug administration is not possible¹⁰⁴

Once eating and drinking

- Give full course of oral antimalarial treatment as per guidelines for uncomplicated malaria
- Avoid mefloquine-containing artemisinin-based combination therapies (ACT) in patients initially presenting with impaired consciousness because of increased risk of post-malaria neurological syndrome

Uncomplicated malaria

Uncomplicated *Plasmodium falciparum*, *Plasmodium knowlesi*, or chloroquine-resistant *Plasmodium vivax* malaria

- Artemether-lumefantrine: 0.83–4.00 mg/kg artemether and 4.83–24.00 mg/kg of lumefantrine given twice a day for 3 days with fatty food; the first two doses should be given 8 h apart
- Artesunate-mefloquine: 4 mg/kg per day artesunate (range 2–10 mg/kg) and 8.3 mg/kg per day mefloquine (7–11 mg/kg) given once a day for 3 days
- Dihydroartemisinin-piperaquine: 4 mg/kg per day dihydroartemisinin (range 2–10 mg/kg) and 18 mg/kg per day piperaquine (16–27 mg/kg) given once a day for 3 days

- for adults and children weighing ≥25 kg; children weighing <25 kg: at least 2.5 mg/kg per day dihydroartemisinin and 20 mg/kg per day piperaquine given once a day for 3 days
- Artesunate-amodiaquine (sensitive *P falciparum* only): 4 mg/kg per day artesunate (range 2–10 mg/kg) and 10 mg/kg per day amodiaquine (7.5–15 mg/kg) once a day for 3 days
- Artesunate-sulfadoxine-pyrimethamine (sensitive *P falciparum* only): 4 mg/kg per day artesunate (range 2–10 mg/kg) given once a day for 3 days and a single administration of at least 25 mg/kg sulfadoxine (25–70 mg/kg) and 1.25 mg/kg pyrimethamine (1.25–3.5 mg/kg) given as a single dose on day 1
- Infants weighing <5 kg with uncomplicated *P falciparum* malaria: treat with ACT at the same mg/kg dose as for children weighing 5 kg
- In the first trimester of pregnancy, use artemether-lumefantrine in preference to the other ACTs

Reducing risk of transmission of *P falciparum* in endemic areas

- Primaquine* single dose of 0.25 mg/kg without G6PD testing (maximum 15 mg base) given concomitantly with blood schizontocidal treatment

Uncomplicated chloroquine-sensitive *P vivax*, *Plasmodium malariae*, and *Plasmodium ovale* malaria

- Chloroquine: 10 mg base per kg day 1 and day 2, 5 mg base per kg on day 3; for children aged <5 years, total dose 30 mg/kg over 3 days

Antirelapse therapy for *P vivax* or *P ovale* malaria

- Primaquine*: 0.25–0.50 mg/kg per day for 14 days given concomitantly with blood schizontocidal therapy and after exclusion of G6PD deficiency; lower risk of relapse in areas of high relapse periodicity if a dose of 0.5 mg/kg per day is used

*Primaquine is contraindicated in pregnant women, infants aged <6 months, and women breastfeeding infants aged <6 months.

severe malaria, are unable to take oral therapy, have any evidence of vital organ dysfunction, or have high parasitaemia should receive intravenous artesunate immediately (panel 2). Intravenous artesunate reduces mortality in adults by 35% and in children by 22% compared with intravenous quinine.^{7,8} If artesunate is not available, intramuscular artemether is a less effective, second-line alternative.¹⁰⁵ Intravenous quinine should only be used as monotherapy if parenteral artemisinin therapy is unavailable. The addition of intravenous quinine to artesunate has been recommended for severe malaria in regions with artemisinin resistance,^{5,10,106} although trial evidence of improved combination efficacy is not yet available.¹⁰ Parenteral antimalarial therapy for severe malaria should be followed by a full course of oral treatment. In general, a minimum of 24 h of intravenous artesunate is recommended before switching to an oral ACT.

Supportive treatments for severe malaria

Immediate supportive treatment addressing relevant manifestations of severe disease should be implemented simultaneously with intravenous antimalarial treatment. Fluid therapy in both children and adults with signs of impaired perfusion without hypotension should be given with caution, preferably crystalloids at a conservative rate.^{107–110} Restrictive fluid therapy is safe and does not worsen kidney function or tissue perfusion in adults with severe falciparum malaria.¹⁰⁸ Aggressive fluid therapy can exacerbate intracranial hypertension¹⁰⁹ and is associated with pulmonary oedema and increased mortality.^{107,110,111} In some patients, small fluid boluses should be given if there are clear signs of shock, which might be caused by concomitant bacterial infection.^{107,110} Immediate whole-blood or red-cell transfusion should be given for severe anaemia. Proposed arbitrary haemoglobin thresholds for transfusion are less than 5 g/dL for children in high-transmission areas and less than 7 g/dL for patients in low-transmission settings.⁸¹ Overall mortality in children with severe anaemia does not differ between those receiving 20 mL/kg of whole blood versus 30 mL/kg.¹¹² In comatose patients, endotracheal intubation and mechanical ventilation for airway protection is indicated where available.¹¹³ Lumbar puncture is indicated in those with altered consciousness to exclude meningitis and assess intracranial pressure.⁵ Lumbar puncture does not increase mortality in clinically stable children with coma, even when brain swelling is present.¹¹⁴ Convulsions in cerebral malaria should be managed similarly to repeated seizures of any other cause.⁵ Phenobarbitone should not be given without respiratory support available.⁵ Renal replacement therapy reduces mortality in severe AKI¹¹⁵ and is indicated in AKI with substantial acidosis, hyperkalaemia or fluid overload, or if multiorgan dysfunction is present.^{5,113}

Invasive bacterial co-infection, frequently caused by *Salmonella* species, is estimated to occur in 5–12% of

children with severe falciparum malaria and is associated with up to a third of deaths in children diagnosed with severe malaria.^{5,116} Many children are misdiagnosed with severe falciparum malaria when they have bacterial sepsis or meningitis with incidental parasitaemia.^{79,117} Rates of bacteraemia in adults hospitalised with falciparum malaria have varied¹¹⁸ from approximately 1% of patients with severe malaria in Viet Nam¹¹⁹ to 15% of adults hospitalised for malaria in Myanmar,¹²⁰ with bacteraemia associated with increased risk of death. Concomitant bacteraemia is difficult to identify clinically in low-resource settings.¹²⁰ Empirical broad-spectrum antibiotics should be given to all children diagnosed with severe malaria, particularly in moderate-to-high transmission settings and should include meningitis cover in those with altered consciousness. Empirical broad-spectrum antibiotics should also be strongly considered in adults diagnosed with severe malaria, ideally after the collection of blood cultures.

Patients with severe malaria should have frequent monitoring of vital signs, level of consciousness, renal function, haemoglobin, and glucose.⁷⁷ Glucose should be replaced to ensure normoglycaemia. Parasitaemia should be assessed every 12–24 h during the initial phase of treatment.¹²¹

Adjunctive therapy for severe malaria

Regular paracetamol, the only adjunctive agent to show benefit in severe malaria, reduces haem-mediated AKI in adults with moderate and severe falciparum and knowlesi malaria, particularly those with substantial intravascular haemolysis.^{53,122} Various adjunctive treatments for cerebral malaria have not shown clinical benefit and should be avoided.⁷⁷ Some, including mannitol¹²³ and phenobarbitone prophylaxis, increase mortality. Dexamethasone increases the risk of gastrointestinal bleeding by a factor of 8.¹²⁴

Uncomplicated malaria

Treatment of uncomplicated malaria depends on the infecting *Plasmodium* species and likelihood of drug resistance. Uncomplicated *P falciparum*, *P knowlesi*, and chloroquine-resistant *P vivax* malaria should be treated with an approved ACT (panel 2; figure 2).⁸¹ ACTs include a rapidly eliminated and potent artemisinin derivative that clears parasitaemia quickly and a more slowly eliminated partner drug that clears residual parasites and reduces selection of artemisinin-resistant clones. Six ACTs are recommended by WHO and artemether–lumefantrine is the most used.¹²⁵ Standard dosing regimens for the lumefantrine component are sub-optimal in young children and pregnant women and should probably be prolonged to improve efficacy in these populations.¹²⁶ Administration of artemether–lumefantrine with fatty food or milk is recommended to optimise absorption.⁸¹ The most recently approved ACT, artesunate–pyronaridine, appears at least as effective as

other ACTs and might have advantages when there is pre-existing parasite resistance to other agents.¹²⁷

Blood schizontocidal regimens active against *P falciparum* are also active against the blood stages of *P vivax*, *P malariae*, and *P ovale*. In general, chloroquine can be used for these species as first-line therapy except where *P vivax* strains have developed chloroquine resistance (panel 2; figure 2).¹²⁸

WHO now recommends artemisinin-based combination therapy (except for artesunate–sulfadoxine–pyrimethamine) for uncomplicated malaria in all trimesters of pregnancy following a systematic review showing fewer adverse pregnancy outcomes in patients treated with artemether–lumefantrine versus quinine in the first trimester.¹²⁹ Artemether–lumefantrine is preferred over other ACTs in the first trimester due to greater accumulated evidence of safety.

Antirelapse therapy for *P vivax* and *P ovale* malaria

In addition to blood-stage therapy, patients with *P vivax* or *P ovale* malaria should receive a hypnozoitocidal drug to prevent future relapses. The 8-aminoquinoline, primaquine, is the primary therapeutic option for antirelapse treatment. Tafenoquine, a new, long-acting analogue of primaquine that can be given as a single dose, was first approved for use in 2018,¹³⁰ but remains inaccessible in most countries with vivax malaria. Tafenoquine might require higher dosing than currently recommended for optimal efficacy and appears to be less active when coadministered with an ACT than chloroquine.^{131,132} Both drugs can cause oxidative haemolysis in patients with G6PD deficiency with the severity of haemolysis proportional to the dose of drug and degree of enzyme deficiency.¹³³ G6PD deficiency should be excluded before prescription of either drug, but this recommendation is often not followed or hypnozoitocidal treatment omitted altogether in endemic regions due to unavailability of suitable assays.¹³⁴ Point-of-care G6PD tests have been developed and trialled,¹³⁵ and are starting to be deployed in some countries.

Primaquine is typically given over 14 days to achieve a total dose of 3·5–7·0 mg/kg. Adherence to this protracted regimen can be poor.^{136,137} WHO endorses a shorter 3·5 mg/kg course over 7 days (0·5 mg/kg per day) as an alternative regimen,⁸¹ but a 7·0 mg/kg total dose, when G6PD activity can be assessed reliably, is more efficacious in most regions¹³⁸ and is the preferred regimen when administered over 14 days.^{139,140} Increasing the dose of primaquine to 1·0 mg/kg per day and shortening the total duration of treatment to 7 days is as efficacious as 0·5 mg/kg per day over 14 days and is likely to improve treatment adherence.¹⁴¹ However, the 1 mg/kg daily dose is associated with gastrointestinal intolerance and risk of severe haemolysis in groups with haemolytic vulnerability (eg, G6PD-heterozygote female patients¹⁴²).¹⁴¹ Children might require higher primaquine doses to achieve optimum therapeutic drug concentrations.¹⁴³

Malaria in pregnancy

Malaria in pregnancy can be fatal to both the mother and fetus. In sub-Saharan Africa, maternal falciparum malaria contributes to 10–20% of maternal deaths and is indirectly responsible for 75 000–200 000 infant deaths.^{73,144} Non-immune pregnant women and semi-immune primigravidae are at particularly high risk of severe malaria.¹⁴⁵ Maternal malaria impairs placental function, affecting fetal development and thereby increasing the risk of miscarriage, low birthweight, small for gestational age, preterm delivery, and stillbirth.¹⁴⁶

In highly malaria-endemic areas where women of childbearing age typically have robust immunity, malaria in pregnancy (particularly in multigravidae) is often asymptomatic but can still cause chronic severe anaemia.¹⁴⁷ Although *P falciparum* causes by far the greatest burden of maternal malaria, adverse maternal and pregnancy outcomes also occur following *P vivax* infections.⁷²

Obtaining a laboratory diagnosis of malaria in pregnant women is challenging. Peripheral blood microscopy might be parasite-negative because of placental parasite sequestration.¹⁴⁷ In highly endemic areas, acquired immunity frequently limits parasitaemia to concentrations below detection thresholds for microscopy and RDTs.¹⁴⁸ Therefore, universal coverage of effective maternal malaria-prevention strategies in endemic regions is crucial.

Congenital and neonatal malaria

Congenital and neonatal malaria is defined as parasitaemia within the first 7–28 days of life.¹⁴⁶ The estimated global prevalence of symptomatic congenital and neonatal malaria is 0·1–0·6%.¹⁴⁹ Failure to include congenital and neonatal malaria as a differential diagnosis in sick neonates presenting with sepsis syndrome in endemic areas can lead to life-threatening treatment delay.^{149,150} Maternal-to-fetal malaria transmission can occur in utero or at delivery.^{151,152} Although most patients are asymptomatic and parasitaemia might spontaneously clear, symptomatic disease can manifest later between 2 and 20 weeks of age.¹⁵³ The incidence of congenital *P falciparum* and *P vivax* malaria is significantly reduced following highly effective maternal malaria treatment.^{146,149,154}

Antimalarial drug resistance

The recurrence of malaria following treatment can be because of incomplete clearance of parasitaemia (recrudescence), reinfection, or, in the case of *P vivax* and *P ovale* malaria, relapse. Recrudescence is more probable if there is initial hyperparasitaemia, suboptimal antimalarial drug dosing or absorption, use of substandard or fake antimalarial drugs, incomplete adherence to treatment, or parasite resistance to the antimalarial drugs used.

Antimalarial resistance has plagued efforts for malaria control since therapeutics were introduced. Chloroquine is no longer active against *P falciparum*, except in central America, and is losing efficacy against *P vivax*

(figure 2). *P. falciparum* resistance to mefloquine emerged on the Thailand–Myanmar border in the 1990s and preceded the use of mefloquine as a partner drug to artesunate.¹⁵⁵

The derivatives of artemisinin, obtained from sweet wormwood (*Artemisia annua*), were developed as pharmaceutical agents in China during the 1970s¹⁵⁶ and introduced in ACTs in the 1990s. Partial resistance of *P. falciparum* to artemisinin, manifesting as delayed parasite clearance, was first confirmed in 2007 in Cambodia¹⁵⁷ and is now widespread across the Greater Mekong subregion.¹⁵⁸ In some patients, pre-existing resistance to an ACT partner drug catalysed selection of artemisinin resistance.¹⁵⁹ Failure of the artemisinin component has likewise accelerated selection of resistance-conferring mutations against partner drugs. Clinical failure by day 28 after artemether–lumefantrine, dihydroartemisinin–piperaquine, and artesunate–mefloquine is now common in Thailand, Cambodia, Viet Nam, and Laos.¹⁶⁰ The molecular signature of artemisinin resistance is a mutated *PfKelch13* gene and polymorphisms causally linked or associated with phenotypic artemisinin resistance have now been detected throughout Asia and in Uganda, Rwanda, Ethiopia, and Eritrea (figure 2).^{161–166}

Although resistance to artemisinins and partner drugs is an important threat to global malaria control, the ACTs remain effective in most parts of the malaria-endemic world. Triple-drug artemisinin-based therapy,^{167–169} artemisinins paired with pyronaridine,¹⁷⁰ and longer treatment courses⁹ have been used successfully for treatment in areas with high-grade resistance and could prolong the useful life of artemisinin-based therapies. Prioritised elimination of malaria from resistance hotspot areas has been promulgated as an important means of limiting further spread of artemisinin-resistant *P. falciparum* strains.

Delayed complications of malaria

Long-term cognitive impairment arises from three of the major clinical syndromes associated with acute mortality in children with falciparum malaria: cerebral malaria,^{171,172} severe anaemia,¹⁷³ and AKI.¹⁷⁴ Other neurological sequelae of cerebral malaria include epilepsy, motor and visual deficits, and short-term developmental delay.^{42,171,175} A reversible, post-malaria neurological syndrome can develop rarely after recovery from severe and uncomplicated falciparum malaria.¹⁷⁶ AKI can lead to chronic kidney disease.¹⁷⁴

Haematological complications include residual severe anaemia at hospital discharge associated with high mortality in ensuing months,¹⁷⁷ preventable with post-discharge malaria chemoprophylaxis. Post-artesunate delayed haemolysis because of the removal of once-infected red cells can manifest as mild-to-moderate anaemia and, rarely, severe anaemia or AKI 7–21 days after artesunate treatment, particularly in those with no immunity and

high initial parasitaemia.^{178–180} Splenic rupture might follow acute vivax or falciparum malaria¹⁸¹ and is more common in adults than children. Chronic infection can cause hyperreactive malarial splenomegaly.²³ Cumulative heavy exposure to *P. falciparum* is associated with risk of Burkitt's lymphoma in African children.¹⁸²

The relationship between malaria and malnutrition is bidirectional. Recurrent malaria causes malnutrition, with malnutrition also increasing risk of severe malaria and death in both falciparum⁵⁰ and vivax infections.¹⁸³ Recurrent and chronic malaria also causes iron deficiency, most likely through hepcidin-mediated block in iron absorption.

Factors that modulate the risk of malaria

Both heritable and acquired factors modulate the risk of *Plasmodium* infection and malaria severity (appendix). Most heritable factors alter either the efficiency of parasite red-cell invasion or intraerythrocytic survival and collectively account for approximately a third of the variability in risk of severe disease.¹⁸⁴ Heterozygotes for HbS (sickle trait) are afforded 85–90% protection against severe and fatal *P. falciparum* malaria, but are not protected against asymptomatic parasitaemia.^{185–188}

G6PD deficiency has considerable geographical overlap with malarious areas.¹⁸⁹ Despite inconsistent evidence, G6PD deficiency probably confers protection against severe falciparum malaria.^{190,191} G6PD deficiency is also associated with protection against uncomplicated vivax and knowlesi malaria.^{192–194}

The Duffy red blood cell antigen, a surface chemokine receptor, is one of two invasion ligands for *P. vivax*. Mutations that prevent expression of the Duffy antigen on erythroid cells provide near-total protection against microscopically detectable *P. vivax* infections and have reached fixation prevalence across most of sub-Saharan Africa.¹⁹⁵

Multiple acquired factors affect the risk of both *Plasmodium* infection and severe disease once infected. Iron deficiency is associated with an up to 45% reduction in the risk of clinical falciparum malaria in children.¹⁹⁶ The COVID-19 pandemic disrupted access to malaria care and preventive activities. Evidence that patients with malaria and COVID co-infection have an increased risk of severe disease is mixed.^{197–199} HIV infection increases *Plasmodium* parasitaemia and malaria severity,²⁰⁰ whereas malaria might increase HIV viral load and disease progression.²⁰¹

Prevention

Chemoprevention

Prompt diagnosis and effective treatment of malaria not only benefits individuals, but is also a cornerstone of malaria prevention. Clearance of parasitaemia with ACTs minimises the risk of onward mosquito transmission primarily by limiting development of further gametocytes. This risk is reduced further by administering a single dose of primaquine to sterilise mature gametocytes in *P. falciparum* malaria.

See Online for appendix

Antimalarial drugs are also used for mass-chemoprevention or targeted-chemoprevention programmes. Seasonal malaria chemoprevention during the high-transmission season, now widely deployed across sub-Saharan Africa in preschool children, has been credited with major reductions in malaria morbidity and mortality.²⁰² In regions with year-round malaria transmission, perennial malaria chemoprevention (formerly intermittent preventive treatment of malaria in infancy⁸¹) reduces incidence of clinical malaria, hospitalisation, and anaemia in children aged <2 years.^{203,204} Provision of three or more courses of sulfadoxine–pyrimethamine during pregnancy (ie, intermittent preventive treatment in pregnancy) is well tolerated and reduces maternal moderate-to-severe anaemia, antenatal parasitaemia, placental parasitaemia, spontaneous abortion, and low birthweight in highly endemic areas.²⁰⁵ In some low-transmission areas outside of Africa, intermittent preventive treatment in pregnancy with dihydroartemisinin–piperaquine is more effective for preventing malaria than screening and treatment at the initial pregnancy visit.¹⁴⁸ Intermittent preventive treatment, particularly of infants, and, to a lesser extent, insecticide-treated bednets, have been estimated to be the most cost-effective interventions to control malaria.²⁰⁶

Vector control

Bespoke engineering solutions that are suitable to local topography and climate have historically been successful in reducing mosquito breeding sites and populations. Current vector-control strategies focus on use of long-lasting insecticide-treated nets and indoor residual spraying to reduce indoor mosquito biting. These strategies have substantially reduced malaria morbidity,^{207,208} particularly in moderate-to-high transmission settings, but are threatened by ongoing evolution of mosquito resistance to pyrethroid insecticides.²⁰⁹ Bednets impregnated with two active insecticidal ingredients with different modes of action provide better malaria protection than single pyrethroid-only nets in areas of pre-existing mosquito pyrethroid resistance.^{210,211} The addition of the insecticide synergist piperonyl butoxide—a substance that blocks enzymatic resistance to pyrethroids within the mosquito—to pyrethroid-treated nets also improves protection.²¹² As well as chemical resistance, important *Anopheles* vector species are increasingly displaying behavioural resistance to standard vector-control measures by biting outdoors and during the daytime.^{213,214} Outdoor residual spraying has shown promise as a potential means of combating this behavioural change in Myanmar.²¹⁵

Panel 3: Priority areas for malaria research and development

Vector control

- Genetic manipulation of mosquitoes to confer refractoriness to *Plasmodium* development and transmission
- New insecticide classes for use in long-lasting, insecticide-impregnated bednets and indoor residual spraying
- Novel strategies for mosquito population control, such as targeted sugar baiting

Diagnostics

- High throughput and very sensitive serological or molecular assays for detecting residual *Plasmodium* transmission (human-only and zoonotic species) in regions approaching elimination
- High throughput assays for detecting *Plasmodium* carriage in mosquitoes
- Ultrasensitive rapid tests for detecting hidden reservoirs of infection: low density circulating infections, placental malaria, and splenic reservoirs
- Rapid diagnostic tests for specifically identifying non-falciparum species
- Tests for hypnozoite carriage
- Practicable and cheap point-of-care tests for G6PD deficiency
- Point-of-care diagnostic for identifying drug-resistant parasites

Therapeutics

- New blood schizonticidal drugs that are active against parasites with artemisinin and partner-drug resistance

- Effect and cost-effectiveness of wide deployment of triple artemisinin combination therapy in preventing or delaying the spread of artemisinin resistance
- Rectal formulations of broad-spectrum antibiotics for combined use with pre-referral rectal artesunate
- Single-dose, combination drug regimens for use in malaria treatment and chemoprophylaxis
- Simplified primaquine regimens for *Plasmodium vivax* radical cure
- Optimised dosing of tafenoquine for improved *P vivax* radical cure
- New hypnozoitocidal drugs that do not cause oxidant haemolysis

Vaccines and monoclonals

- Effect of deployment of pre-erythrocytic vaccines with improved efficacy
- Effect and cost-effectiveness of monoclonal antibody immunoprophylaxis in different transmission settings and pregnancy
- Transmission-blocking vaccines
- Vaccines for *P vivax* malaria

Data

- Improved timeliness, cohesiveness, granularity, and accessibility of malaria incidence and resistance surveillance data
- Expanded use of mobile phone technology for patient education, treatment supervision, and malaria surveillance

Anopheles stephensi has recently spread from Asia to the Horn of Africa.²¹⁶ This highly efficient vector of both *P falciparum* and *P vivax* prefers breeding in human-made containers and urban environments and thus poses a substantial risk of increased receptivity to local transmission in urban Africa.

Vaccines and monoclonals

Malaria vaccine development includes four main concept categories: (1) blood-stage vaccines to control erythrocytic merozoite multiplication; (2) pre-erythrocytic vaccines to prevent blood-stage infection altogether; (3) placental malaria vaccines targeting placenta-sequestering parasites; and (4) transmission-blocking vaccines that kill sexual or mosquito stages.²¹⁷ Of these categories, pre-erythrocytic vaccines are the most advanced. Despite modest efficacy, RTS,S/AS01E, a pre-erythrocytic vaccine, was recommended by WHO in 2021 for widespread use in young children in areas of moderate and high *P falciparum* transmission—the first malaria vaccine to receive such endorsement.²¹⁸ In African phase 3 clinical trials, three doses of the vaccine administered 1 month apart followed by a booster dose 18 months later reduced clinical malaria episodes over 3–4 years by 36% in young children first vaccinated between 5 months and 17 months of age and 26% in infants first vaccinated between 6 months and 12 weeks of age.²¹⁹ Protection against severe malaria was 32% in young children and 26% in infants. R21, another pre-erythrocytic, CSP-based vaccine, has potential for improved efficacy.²²⁰

The combination of chemoprophylaxis and vaccination holds greater promise than vaccines alone. Booster doses of RTS,S before the wet season in combination with seasonal malaria chemoprophylaxis achieved impressive reductions in clinical malaria, severe malaria, and malaria-related death in west Africa.²²¹

Vaccine development for *P vivax* lags behind that of *P falciparum*, but has followed similar strategies, showing promise in phase 1 studies.²²² Long-acting monoclonal antibodies targeting CSP are efficacious in preventing falciparum malaria in human volunteers²²³ and endemic regions,²²⁴ and raise the potential for single-dose immunoprophylaxis in non-immune travellers and seasonal immunoprophylaxis in endemic areas.

The future of malaria control and elimination

Launched in 1955, the Global Malaria Eradication Program ended without success 14 years later.²²⁵ Since 1969, there has been no further time-limited commitment to global malaria eradication and its feasibility continues to be debated. However, malaria elimination—defined by WHO as the interruption of local transmission of a specific malaria parasite species in a defined geographical area due to deliberate activities²²⁶—is being pursued vigorously in multiple countries with unstable malaria transmission. The Sustainable Development Goals adopted by WHO in 2016 call for a reduction in malaria incidence and

mortality of at least 90% and elimination of malaria in at least 35 countries by 2030. The E-2025 initiative, launched by WHO in 2021, aims to eliminate malaria in 25 countries primarily across Asia and Latin America by 2025.¹ Progress has been uneven, with estimated incidence of malaria rising by over 50% in two African countries and four South American countries targeting elimination,¹ and significantly in several high-burden countries in Africa in the past 5 years. Overall progress against malaria was already deteriorating before the COVID-19 pandemic. Social upheaval from natural disasters, epidemics, and civil unrest (eg, Venezuela²²⁷ and Myanmar²²⁸) have contributed to the resurgence of malaria in some locations. Stagnant funding, substantial gaps in prevention and treatment activities, and worsening antimalarial and insecticide resistance have also contributed to varying degrees.¹ Imprecision of country-level estimates of malaria burden because of suboptimal surveillance data collection and changes in methodology used to estimate incidence figures might also have resulted in inaccurate trends in some locations.²²⁹

Consistent application of existing malaria control tools might be sufficient to achieve elimination in countries that already have low malaria incidence and socioeconomic stability, but meaningful reductions in malaria morbidity in high-transmission settings will require increased political commitment, consistent funding, and novel strategies (panel 3). Malaria surveillance must become more granular, timely, and user-friendly to enable rapid detection and response to hotspots of transmission. Novel vector-control strategies that reduce human labour and overcome changing mosquito behaviour and resistance are needed. The propagation of genes conferring sterility or immunity to *Plasmodium* parasites through mosquito populations (ie, the so-called gene-drive process) and attractive mosquito baits are two promising strategies. New blood schizontocidal drugs that are active against artemisinin-resistant strains are required. Wide deployment of rapid and practicable tests for G6PD deficiency would enable a much broader use of primaquine or tafenoquine for relapse prevention in vivax malaria.

Competing interests for malaria-control resources will continue to arise, dictating greater efficiency of regulatory processes for research, development, and malaria interventions. Improved coordination of malaria control with other communicable disease programmes (eg, in mass drug-administration campaigns) is imperative. Although a malaria-free world is possible, navigating the path to this lofty goal will require original thinking and resolute perseverance.

Contributors

JRP, NMD, NMA, and SK performed the literature search and wrote the first draft of the manuscript. SK prepared the figures.

NMD prepared the antimalarial drug-resistance maps. JRP, NMD, and NMA prepared the panels. DA reviewed sections relevant to the African context. All authors reviewed and approved the final version of the manuscript.

Declaration of interests

We declare no competing interests.

Acknowledgments

We thank our many colleagues and collaborators. The Timika Research Facility and Papuan Health and Community Development Foundation are supported by the Australian Department of Foreign Affairs and Trade. This Seminar was supported in part by the Australian National Health and Medical Research Council (Senior Principal Research Fellowship [1135820] granted to NMA and the Australian Centre for Research Excellence on Malaria Elimination [1134989]).

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